

UNIT-4

→ Device-to-Device (D2D) communications:

- Extension of 4G D2D standardization to 5G
- Radio resource management for mobile broadband

D2D

- multihop & multi-operator D2D comm^{ns}

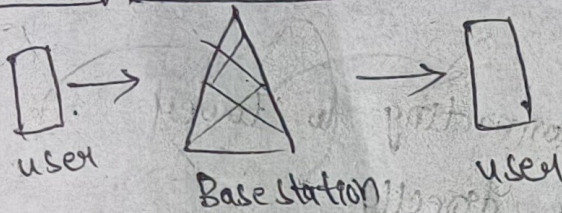
UNIT-IV

* Device-to-Device (D2D) Communication

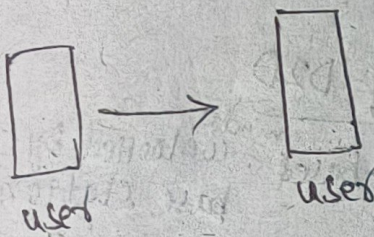
→ D2D Commⁿ is a technology

- where two user devices communicate directly without routing through base station (BS)

Normally / traditional :



In D2D:



→ Reduces delay

→ Improves efficiency

(direct)

Types of D2D

①. Inband D2D (inside n/w)

→ uses mobile n/w

→ but tower controls it

②. Outband D2D (outside n/w)

→ uses Wi-Fi / Bluetooth

→ no tower control

Real-life examples

→ Wi-Fi

→ bluetooth

→ Nearby file sharing

Why?

- reduce n/w load on base station
- Fast commⁿ → improves speed
- Saves Power → Works in poor n/w areas
- useful in emergency situations
- Enables IoT

Ex: ⇒ Sharing file using Wi-Fi Direct

→ In a crowded stadium:

- instead of all users connecting to tower
- Nearby phones communicate directly

→ Direct commⁿ b/w two nearby devices

→ does not require base station

→ reduces n/w traffic

→ Introduced in LTE (4G)
improved in 5G

→ Works in licensed or unlicensed spectrum.

→ Can control:

- Network (Autonomous)
- without Network

→ improves spectral efficiency

→ low latency commⁿ

→ direct commⁿ b/w devices

Adv

- low latency
- high data rate
- reduce load

dis

- interference
- Security & Privacy
- complex

Mode Models of D2D

①. Network-based mode - controlled by base station

②. Autonomous Mode

↓
devices communicate independently

Applications

→ Vehicle-to-Vehicle commⁿ

→ Smart ~~network~~ cities

→ IoT

→ online gaming

→ AR/VR

→ Emergency commⁿ

* Extension of 4G D2D Standardization to 5G

↳ Evolution of LTE D2D to 5G D2D

- 5G improves the limitations of 4G D2D as ProSe
- In 4G LTE, D2D commⁿ was introduced, with limited capabilities

In 4G: (Basic D2D) 4G/LTE D2D

- High delay
- Supports fewer devices
- Limited Speed
- Limited Scalability
- Limited device discovery

ProSe - Proximity Services

In 5G: (Advanced D2D)

- Very high Speed
- Ultra-low latency
- Supports IoT
- More reliable

ex: Autonomous Vehicle

Working of D2D

1. Step-1: Discovery

- Devices find nearby devices

Step-2: Connection

- Devices connect

Step-3: Data Transfer

- Data is sent directly

(RRM) * Radio Resource Management for D2D

- RRM = efficient allocation of radio resources (Power, frequency, time)
- ⇒ Managing frequency, Power, time
- It avoid interference
- Maintain QoS (Quality of Service) → improves efficiency
- Efficient Spectrum usage
- Improves Performance

ex

- ① Two nearby users using same frequency → interference
⇒ RRM assign different time / ~~slot~~ frequency
- ② If 2 users use same signal → Problem
→ So, it gives different time / frequency.

⇒ RRM is the process of ~~an~~ efficiently allocating frequency, time & Power resources to D2D users.

Techniques

- Power control → adjust transmission Power.
- Resource allocation → assign time slots
- Scheduling Algorithms → manage multiple users

Users 1 → Slot 1

Users 2 → Slot 2

⇒ RRM ensures :

- Low interference
- High throughput
- Efficient Spectrum usage.

→ It is the process of allocating & controlling radio resources (frequency, time, power)
- Specially for high speed D2D commⁿ

→ handle interference
Features

- High ~~rate~~ data rate requirement
- Continuous data transmission (video streaming)
- Requires low latency
- High bandwidth

⇒ RRM manages:

- Frequency Resources

- Allocates different frequency bands

- Time Resources

- Assigns time slots

- Power Resources

- controls transmission power

- Spatial Resources

- Uses beamforming, MIMO

- Centralized RRM → Controlled by BS

- Distributed RRM → devices control themselves

Types of RRM

(ex) In a stadium:

- Many users stream videos

RRM ⇒ Allocates different frequency

⇒ assign time

⇒ control power

* Multi-hop D2D Communication → Data travel through multiple devices

→ It is a technique.

- where data is transmitted from source to destination through multiple intermediate devices (relay nodes)

- instead of direct commⁿ.

⇒ In normal D2D:

- Devices communicate directly (Single hop)

→ But when distance is large:

- Direct commⁿ is not possible.

Solⁿ: Multi-hop D2D

⇒ Data is transmitted through intermediate devices (relay nodes)

Need?

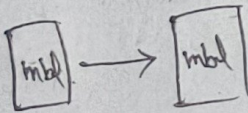
→ Devices are far apart → Direct commⁿ not possible

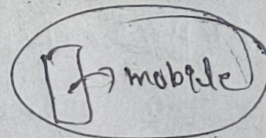
→ Weak signal

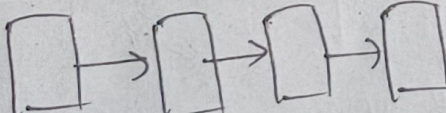
→ No base station

→ Required in: Rural areas, Disaster zones

①

Instead of:  (direct)



Use: 

②

Instead of: $A \rightarrow E$

Use: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$

⇒ Each node acts as relay

Working

1. Source device sends data to nearest device.
2. That device acts as a relay
3. device forwards to next.
4. Process continues.
5. data reaches destination.

Types $\Rightarrow$ RCM

①. Relay-based Commⁿ: (relays)

②. Cluster-based commⁿ

- Devices form groups (clusters)
- cluster head forwards data

③. Mesh-based commⁿ

- Multiple paths available
- Data can take best route

Source



Relay 1



Relay 2



Destination.

Challenges

- Increased delay
- Battery consumption
- Complex
- Security issues

Applications

- Disaster recovery commⁿ
- Military commⁿ
- IoT
- Smart cities

* Multi-operator D2D Commⁿ

- It is a technique.
- Commⁿ b/w devices belonging to different n/w operators.

⇒ In normal D2D:

- Devices belong to same n/w operator

→ But in real life:

- Users belong to different n/w's (Jio, VI, Airtel)

(Solⁿ) ⇒ So, we need Multi-operator D2D

→ It allows direct commⁿ b/w devices of different operators.

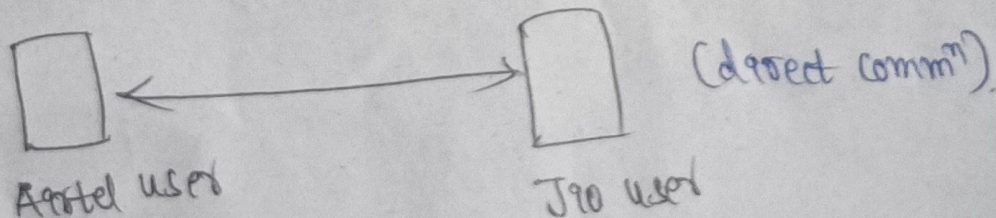
→ Technique where:

- devices belonging to diff mobile n/w operators communicate directly without routing.

(Need)

→ Users use from diff n/w

→ require in emergency, Disaster mgmt.



→ Cross n/w commⁿ

→ Needs:- Security
- Authentication

→ Uses 3GPP

→ Uses standard protocols.

Adv.

- Improves connectivity
- Enable commⁿ across all n/w^s
- Useful in emergencies

Challenges

- Security, Privacy
- Billing & charging
- Complex

Applications

- Disaster recovery
- Military commⁿ
- Smart city
- IoT n/w